

Hantavirus Infection

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Hantavirus infection is a viral disease that is spread from rodents to people. The virus can cause severe infections of the lungs (with cough and shortness of breath) or kidneys (with abdominal pain, and sometimes kidney failure).

- Hantaviruses are spread through contact with infected rodents or their droppings.
- The infection starts with sudden fever, headache, muscle aches, and sometimes abdominal symptoms, which may be followed by a cough and shortness of breath or by kidney problems.
- Blood tests to identify the virus can confirm the diagnosis.
- Oxygen and medications to stabilize blood pressure are used if the lungs are affected, and dialysis may be needed if the kidneys are affected.

(See also [Overview of Arbovirus, Arenavirus, and Filovirus Infections](#).)

Hantaviruses infect various species of rodents throughout the world. The virus is present in the urine and feces of the rodents. The infection is spread when people have contact with rodents, their droppings or urine, or possibly when they inhale virus particles in places with large amounts of rodent droppings. Most hantaviruses do not spread from person to person; rarely, Andes hantavirus in southern South America spreads directly between people in close physical contact. Hantavirus infections are becoming more common.

There are several species of hantavirus. Depending on the species the virus affects different organs:

- The lungs, causing hantavirus pulmonary syndrome (HPS)

- The kidneys, causing hemorrhagic fever with renal syndrome (HFRS)

However, many symptoms of the 2 infections overlap.

The **pulmonary syndrome** has been found in the western United States and Canada as well as countries in Central and South America.

The **renal syndrome** occurs primarily in parts of Europe, Korea, China, and Russia. This virus is spread by brown Norway rats and has been spread throughout the world by rats on ships. A few cases of HFRS hantavirus infection, spread by pet or laboratory rats, have occurred in the United States, Canada, and Europe.

Symptoms of Hantavirus Infection

Symptoms of hantavirus infection begin with sudden fever, headache, and muscle aches, typically about 2 weeks (but possibly as long as 6 weeks) after exposure to the rodent droppings or urine. People may also have abdominal pain, diarrhea, or vomiting.

These symptoms continue for several days.

Hantavirus pulmonary syndrome

People with pulmonary syndrome then develop a cough and shortness of breath, which may become severe within hours. Fluid collects around the lung, and blood pressure becomes low.

The pulmonary syndrome causes death in up to about 50% of people. Those who survive the first few days improve rapidly and recover completely in about 2 to 3 weeks.

Hemorrhagic fever with renal syndrome

In some people with hemorrhagic fever with renal syndrome, the infection is mild and does not cause symptoms.

In others, vague symptoms (such as a high fever, muscle aches, headache, and nausea) begin suddenly. People with mild symptoms recover completely.

In others, symptoms become severe. Very low blood pressure (**shock**) develops in a few people. **Kidney failure** develops, and urine production may stop (called anuria). People may have blood in their urine and/or stool and bruises on their skin. Death occurs in up to 15%, depending on the strain of the virus and a person's underlying

medical problems. Of those who survive, most recover in 3 to 6 weeks, but recovery may take up to 6 months.

Diagnosis of Hantavirus Infection

- Blood tests to identify the virus

Hantavirus infection is suspected when people who may have been exposed to the virus have characteristic symptoms.

Blood tests to identify the virus can confirm the diagnosis.

Doctors do other blood tests to evaluate the function of the kidneys and other organs.

If the pulmonary syndrome is suspected, a chest x-ray may be done.

Echocardiography (ultrasound of the heart) is usually done to exclude other cardiac causes of fluid around the lungs.

Treatment of Hantavirus Infection

- Supportive care
- For the pulmonary syndrome, oxygen and medications to stabilize blood pressure
- For the hemorrhagic fever renal syndrome, dialysis and the antiviral medication ribavirin

Treatment of hantavirus infection is mostly supportive.

For the **pulmonary syndrome**, oxygen and medications to stabilize blood pressure appear to be most crucial to recovery. Sometimes a ventilator is needed to help with breathing or in very severe cases blood oxygenation machine treatment (extracorporeal membrane oxygenation [ECMO]) may be necessary.

For the **hemorrhagic fever renal syndrome**, dialysis may be needed and can be lifesaving. The antiviral medication ribavirin, given intravenously, may help reduce the severity of symptoms and the risk of death.

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the content of these resources.

Centers for Disease Control and Prevention (CDC): [Hantavirus: About Andes Virus](#)

World Health Organization (WHO): [Hantavirus Fact Sheet](#)

Drug Information for the Topic



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